BIKESHARE PROJECT - REPORT

Business Task

After a successful previous year for the **Cyclistic bike-share*** Company, this report wishes to examine the business task for the proposed theory on the orientation of the company's new marketing plan.

Handling more than 5,800 bicycles and 600 docking stations, **Cyclistic**, a company in Chicago, will need to adjust its new marketing strategy. The company has two distinct categories of clients: casual riders and annual members. The customers who purchase single-ride or full-day passes are considered Casual riders. Customers who purchase annual memberships are considered as Members. The company's finance analysts have concluded that annual members are far more profitable; consequently, the director of the marketing department has set the clear goal of designing marketing strategies aimed at converting casual riders into annual members.

In order for the marketing strategy to be effectively designed, the behavior of these two different categories of customers must be analyzed. Therefore, this analysis will be focusing on the following business task:

'How do annual members and casual riders use Cyclistic bikes differently? How could the company's strategy be affected by the findings?'

***Cyclistic** is a fictional company. The data is used for the purposes of this case study under the [Data License Agreement | Divvy Bikes](). The data has been made available by Motivate International Inc.*

Data Sources used

The data that will be analyzed will be the company's historical trip data for 2023, in the city of Chicago. The data can be downloaded from the address: https://divvy-tripdata.s3.amazonaws.com/index.html. More specifically, the dataset consists of 12 CSV files, one for each month of 2023. Since this is first-party data, it is considered credible and has high integrity.

Characteristics of Data

- We do not have general or demographic data on the users but only on the rides. We have the type of membership (member or casual), the type of bike (classic, electric), the time that the ride started and ended, the name of the stations that the ride started from and ended (street), and their respective locations (latitude and longitude).
- We do not know the distance traveled per ride but we know the duration of the ride.
- There are some minor NA values and nulls in the data.

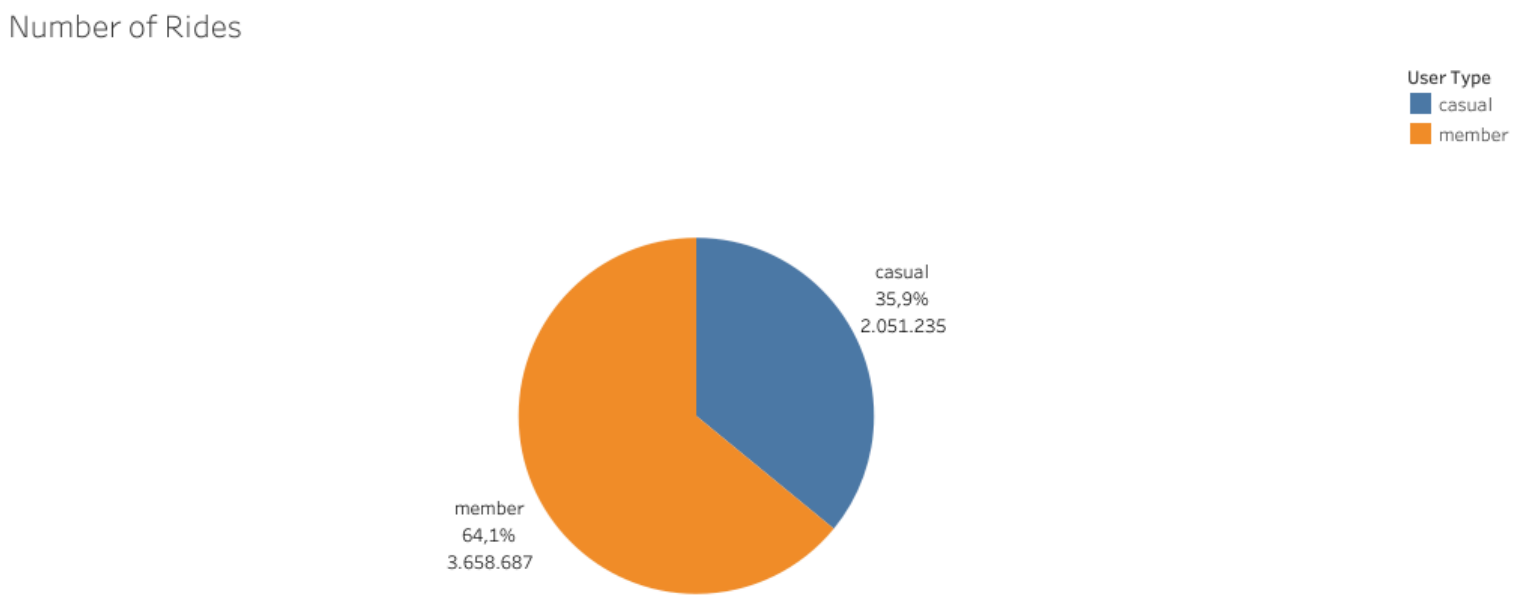
Data Processing and Analysis

In order for the analysis to be accurate, a targeted approach had to be followed, based and oriented on the business task. The question of the business task is focused on the difference in the use of the Cyclistic bikes, between casual riders and members, in order for a marketing plan to be strategized, which would facilitate the conversion of casual users to members.

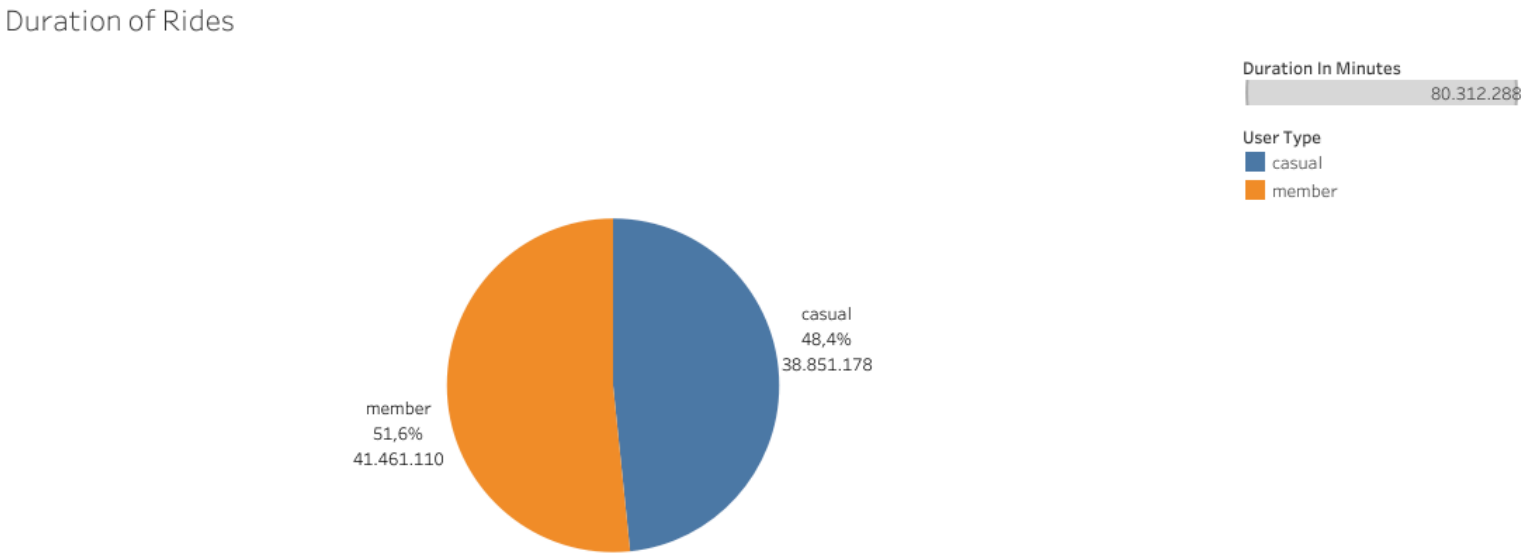
As a result, the data cleaning and subsequent analysis needed to focus on the habits of the members as well as the habits of the casual users; the most important points (based on the existing data) revolve around the duration of time that each user used the service for, alongside the hours, days and months during which the service is mostly used. Consequently, the analysis and the charts that follow examine these metrics in detail.

*The exact cleaning process can be found in the ‘Datalog and Information file’

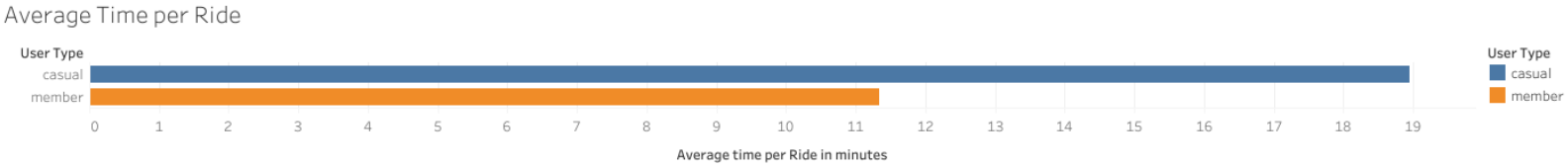
Presentation of findings - Habits of Members vs Casual users



The company has a very dedicated and active member base. Almost two-thirds of the rides are those of members. However, as it can clearly be seen that many casual riders enjoy the services of **Cyclistic bike-share** and the company's aim is to manage and convert them into members, the habits and characteristics of these riders will be thoroughly examined below, based on the data received by the company.

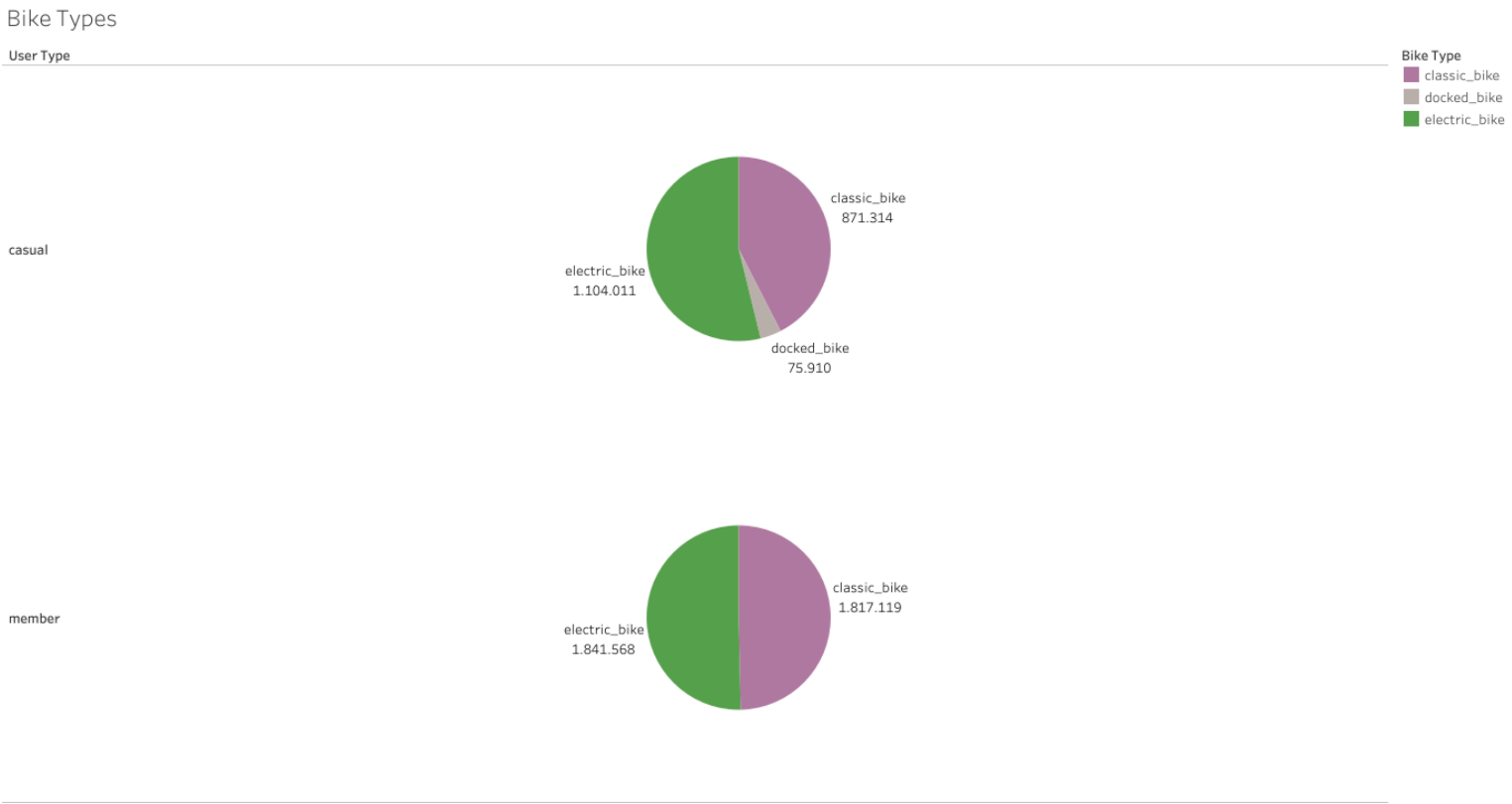


The importance of the casual users can easily be seen in this chart. Even though casual rides account for approximately 35% of the number of all rides, the time spent on these rides constitutes almost half of the total time of all the rides cumulatively.

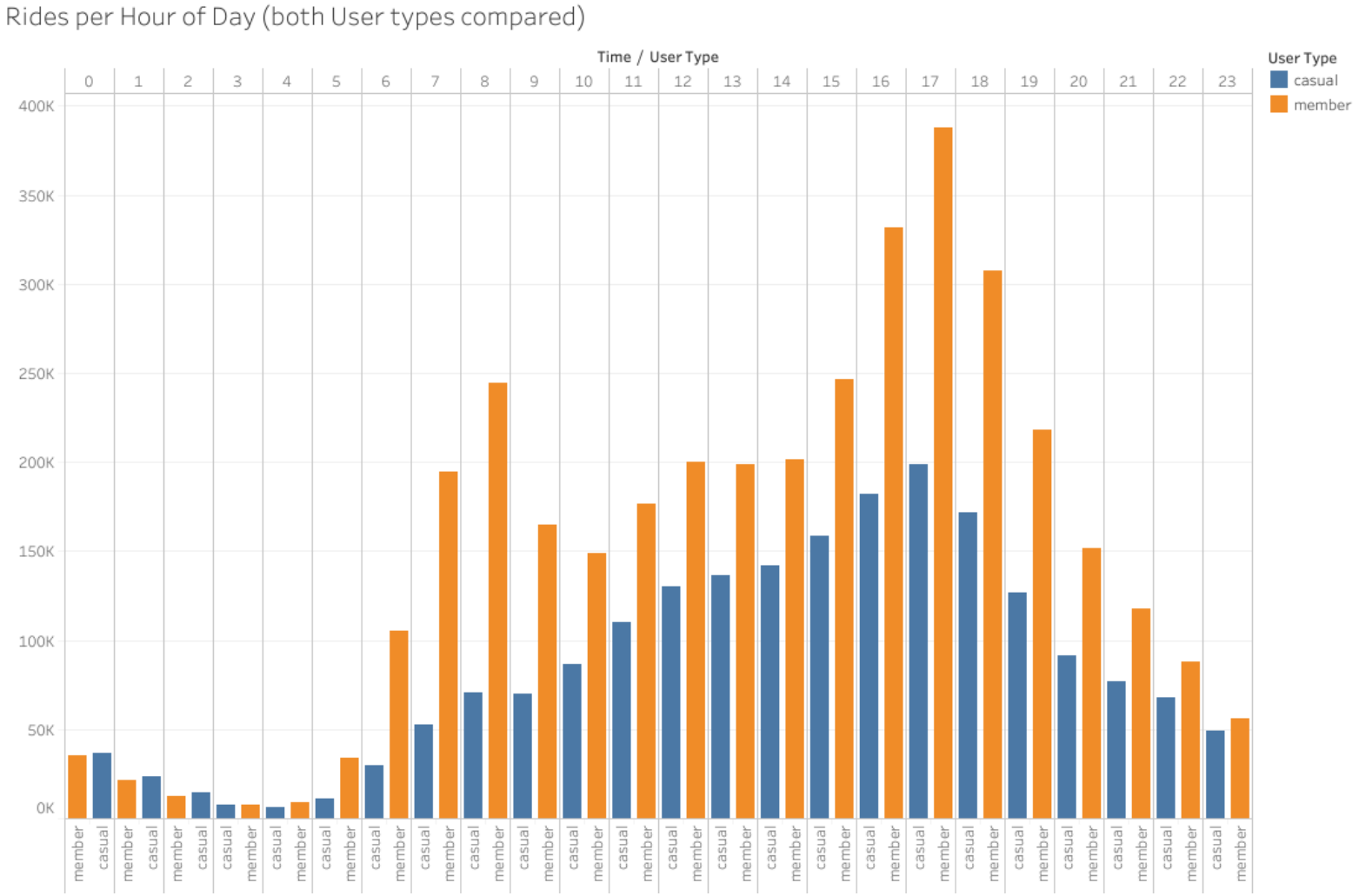


As a result of the two previous charts, the average individual times can be seen per ride, both for casual riders and members. This gives us a first important perspective on the habits of the two types of users; the casual

user uses the bike for a longer period, probably for recreational rides. The member uses the bike for shorter, probably more ‘standard, planned’ rides, as a substitute for a car, to facilitate their movement from point A to B.

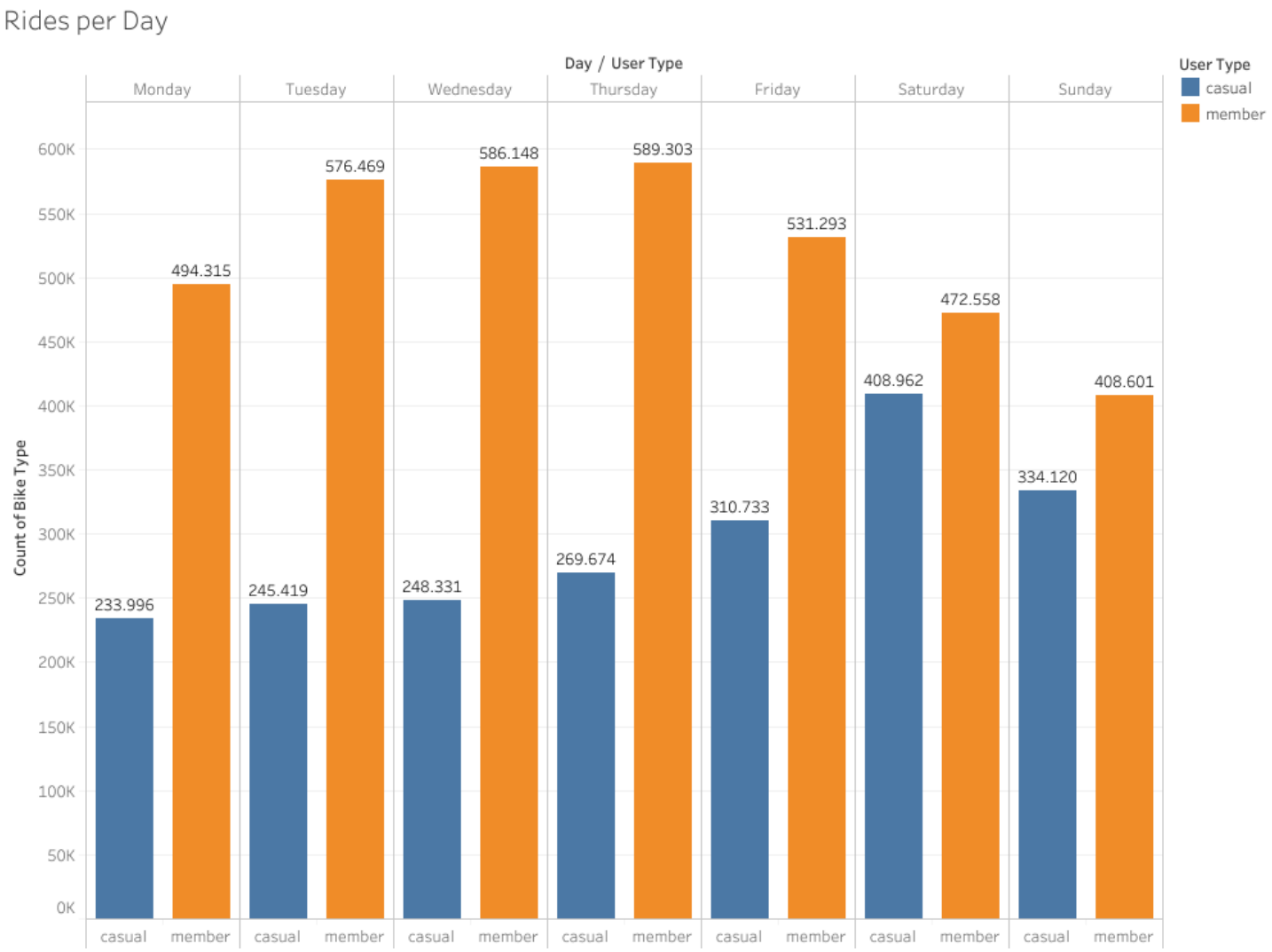


These charts depict the types of bikes that both casual users and members prefer. It can clearly be seen that both classic, conventional bikes and electric bikes are widely used, and with the same frequency, by both groups. As a result, it would not be necessary to focus exclusively on a specific type of bike, marketing-wise. Casual users do show a very slight preference for electric bikes, not necessarily a significant one. There could be a future marketing campaign promoting electric bikes so that more data can be collected and analyzed, regarding the Casual riders’ preferences.



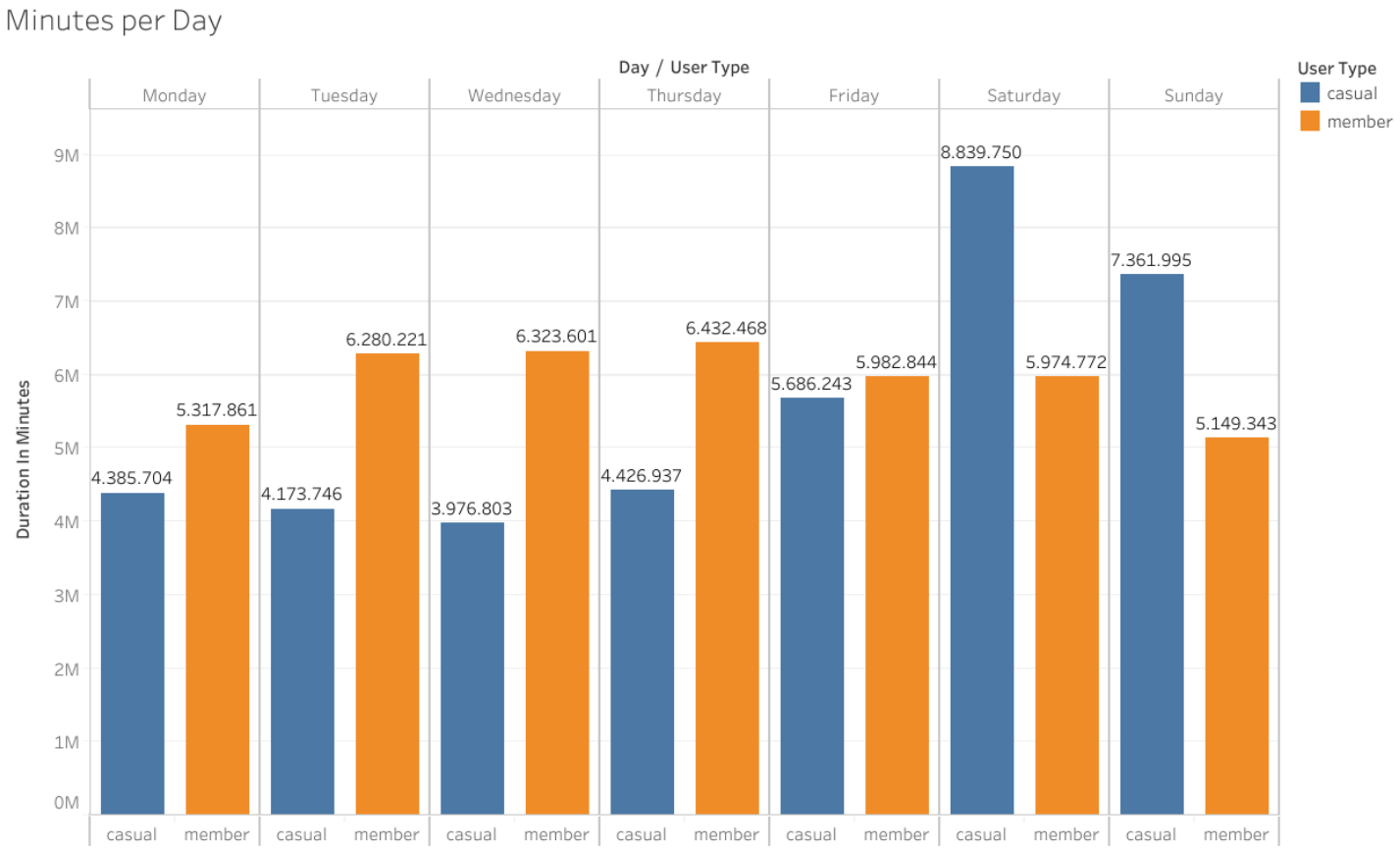
This chart gives us a solid perspective on the habits of both categories of users, regarding the most ‘popular’ time of the day. We can see that the use peaks for both categories at 5 pm, as well as the hours just before and after it. This is the time that people return from work, so it is safe to assume that many users, both members and casual, take the bicycles from their workplace to return home.

This is even more apparent in the members category, the habit of which clearly coincides with the office working hours. Apart from the 5pm peak we can see another smaller, but quite significant one, at 8 am (alongside 7 am and 9 am as well). This indicates that members have a very consistent habit of basing their entire commute (to a great extent) back and forth from work on **Cyclistic bike-share.** The case is not the same in the casual users category, the habits of which indicate that the bikes are used more 'casually' indeed, peaking in the afternoon and early evening. This juxtaposition illustrates even more the solid use of the bikes by the members category as a means of transportation, preferable to cars or mass transit. This will be revealed in the following chart once again, from the perspective of the workdays vs the weekend.



The Rides per Day chart differentiates the two groups, revealing some clear-cut characteristics. While the two groups are very close to the numbers of rides for the weekends, the bike use on the weekends for the Member group is the low point, whereas for the Casual group, it is the exact opposite. That means that members clearly use the bikes more on weekdays, whereas casual users use them slightly more on the weekends.

After this chart, we can now have a more complete perspective, reinforcing the theory that the members use the bikes as a ‘tool’ which is more connected to their work, whereas the casual users as a leisure activity, to be enjoyed more during their free time. However, at this point, it must be highlighted that while the numbers that have been (and will be) examined show some clear preferences and habits of the groups, the rest of the numbers are not negligible. As an example taken from the chart above, the fact that the casual users use the bikes mostly on the weekends does not mean that there is no substantial use of the bikes during the weekdays for that group as well. These facts should also be taken into consideration during the design of a complete marketing policy.

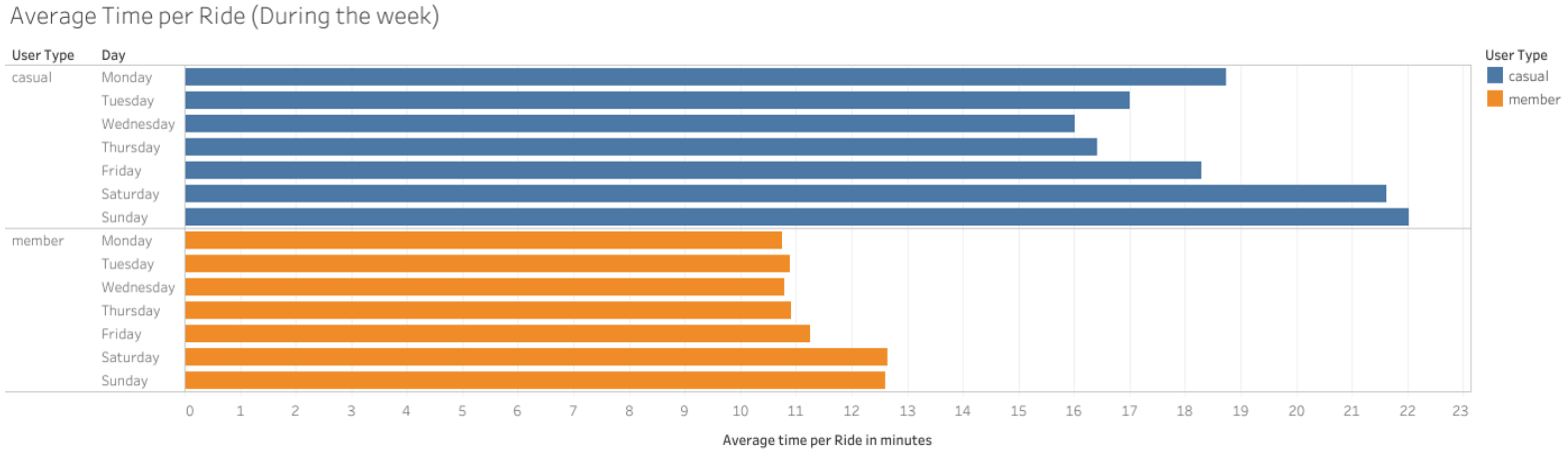


Despite having a clear picture of the habits of both groups by now, this chart offers an even more substantial insight, focusing on the actual time that the users spend riding the bikes. It could even be argued that this is the most useful insight, showing the actual usage of the service (in juxtaposition to the number of rides).

Even though the general metrics are on par with those of the number of rides per day we examined above, the importance of the weekends should be highlighted, for both groups. For the Casual group, the time on both weekend days is significantly higher, especially in the case of Saturdays, on which casual users spend practically twice the time they do during a weekday. Fridays are also more popular for Casual users than the rest of the working days, but again, not as popular as the weekend. All that reinforces the theory that, although many Casual users use **Cyclistic**

during working days and probably for transport to and from work, there is a clear preference for cycling in their free time.

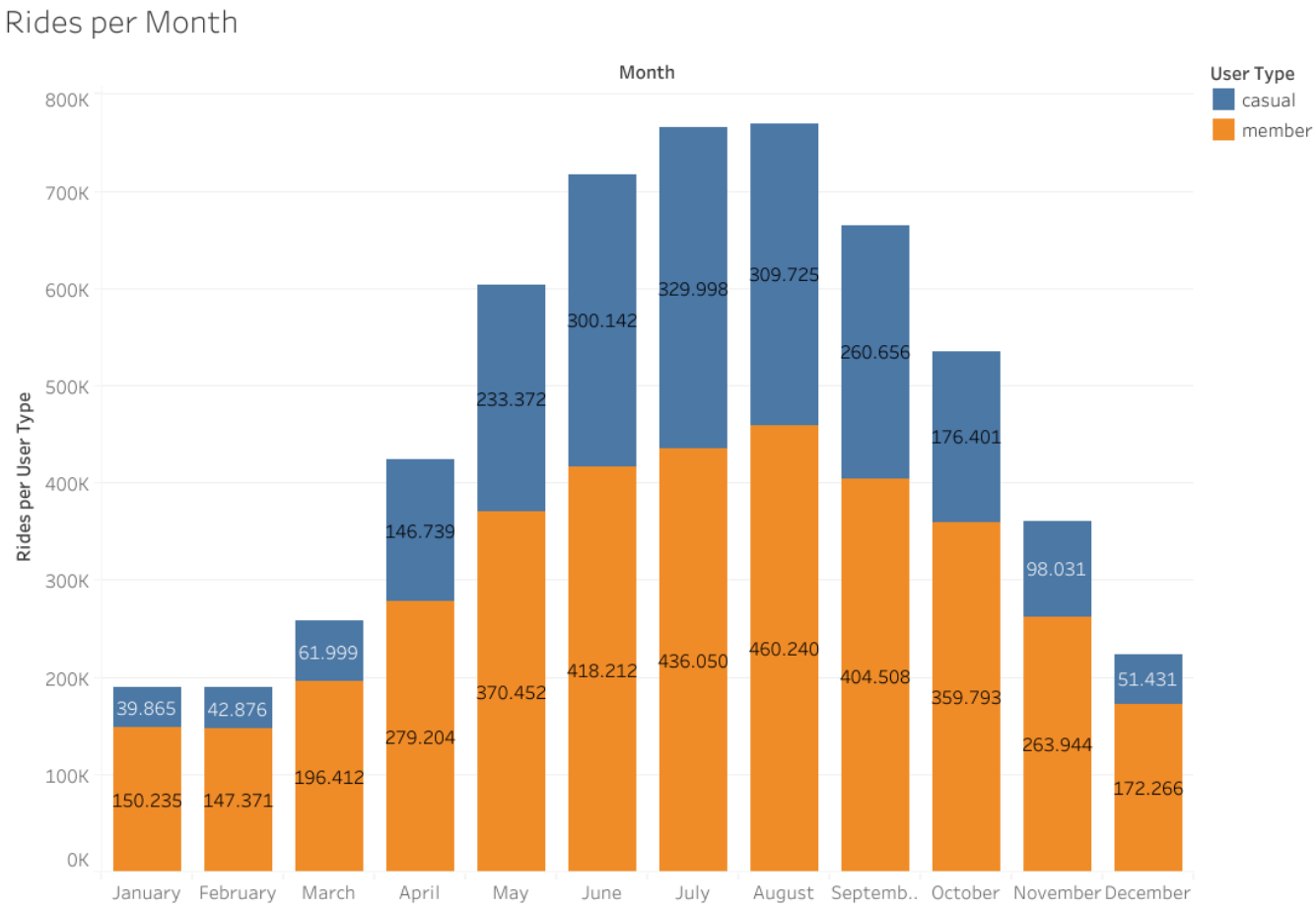
When it comes to the Member group, even though the data follows that of the Number of Rides chart, the discrepancy between the weekdays and the weekend is less noticeable. That means that even though Members use the bikes on the weekdays, for transportation to and from work, they also use them during the weekends as well. While Sunday is the day that the Members use the bikes for the least time, Saturday is almost equally popular as Tuesday to Friday (and even more popular than Moday). From the data above we can safely conclude that even though Members use the service as a tool for their transportation, they also dedicate time on the weekends for recreational purposes and, as we can assume at this point, their rides are even longer, something that will be examined by the next chart.



As it has already been established, the time that users spend every day on the service differentiates the two groups. Again, the Members group shows a very ‘consistent’ allocation of time, since for every weekday the average

Member’s ride will be around 11 minutes. As far as the weekends are concerned, the use will be for around 12 and a half minutes per ride. We can see here that Members will indeed go for slightly longer rides on the weekend.

When it comes to Casual users, we can see that for every given day their rides are longer. The average time per user ride on weekdays ranges from 16 to nearly 19 minutes. For the weekend the average time is close to 22 minutes per ride. Once more, the data complies with the theory that Members have more ‘fixed’ and standard rides while the Casual users spend more time using the bikes for leisure, in a more ‘relaxed’ way.



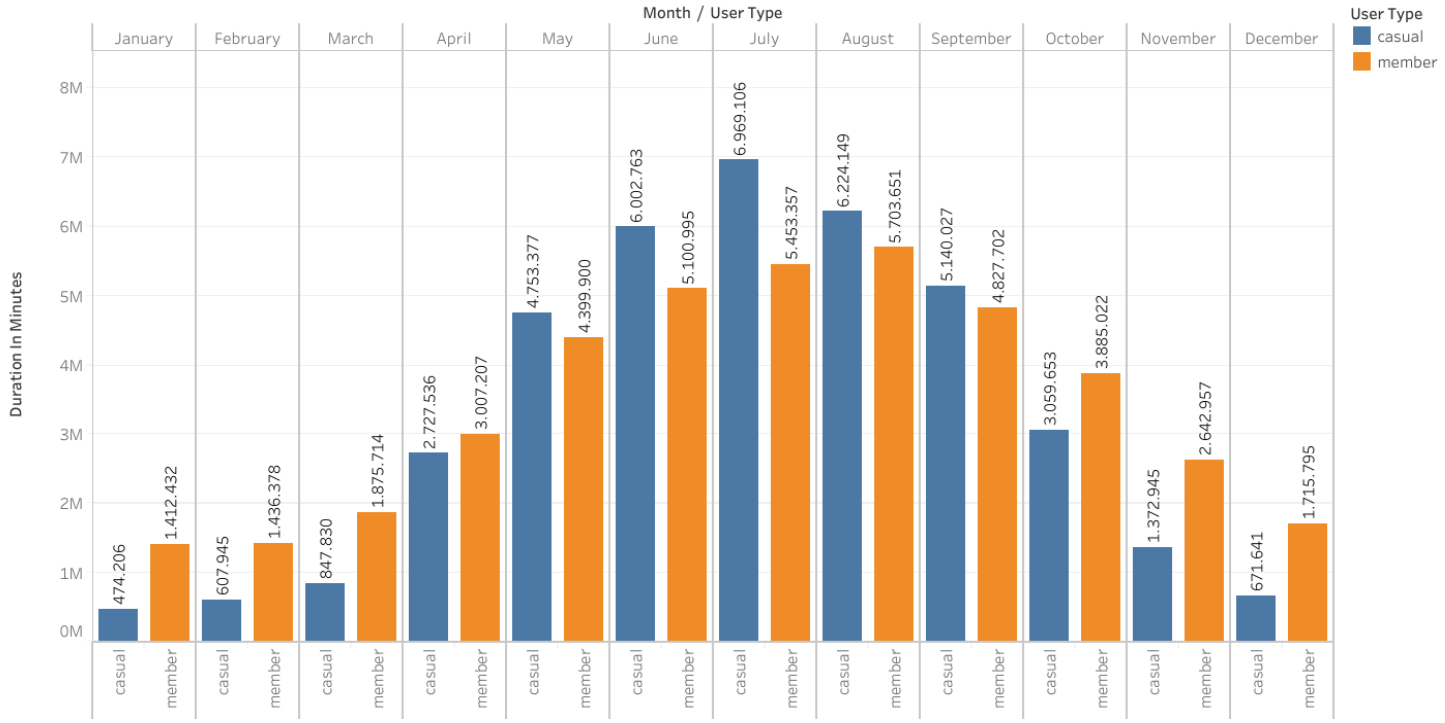
The above chart shows the total rides for both categories, stacked together. As would be expected for the city of Chicago, there is a vast difference in the number of rides for the summer months compared to the winter ones. July and August are the top months, with June following closely, while, on the other hand, January and February are the months with the least rides, with December very close.

Even though both groups have the exact same tendency as far as the busy and less busy months are concerned, an interesting remark is that during the winter months, Casual rides are very limited. There is a very large difference in the busiest months for Casual rides in comparison to the less busy ones. That is not exactly the same in the Members group, since the

discrepancy between the busiest and the less busy months is much less noticeable (but still significant).

The conclusion drawn from this data supports again the ‘recreational character’ of the Casual group, since the winter months are not popular. At the same time, the opposite can be said for the Members group, where a slight but clear consistency can be noted. Even though it could not be expected for the winter months to be as busy as the summer ones, we can see that Members still prefer using the bikes to a significant extent, since they seem indeed to be helpful for them, compared to other means of transport, to and from work. A similar tendency for all data should be expected on the ‘Minutes per Month’ chart as well.

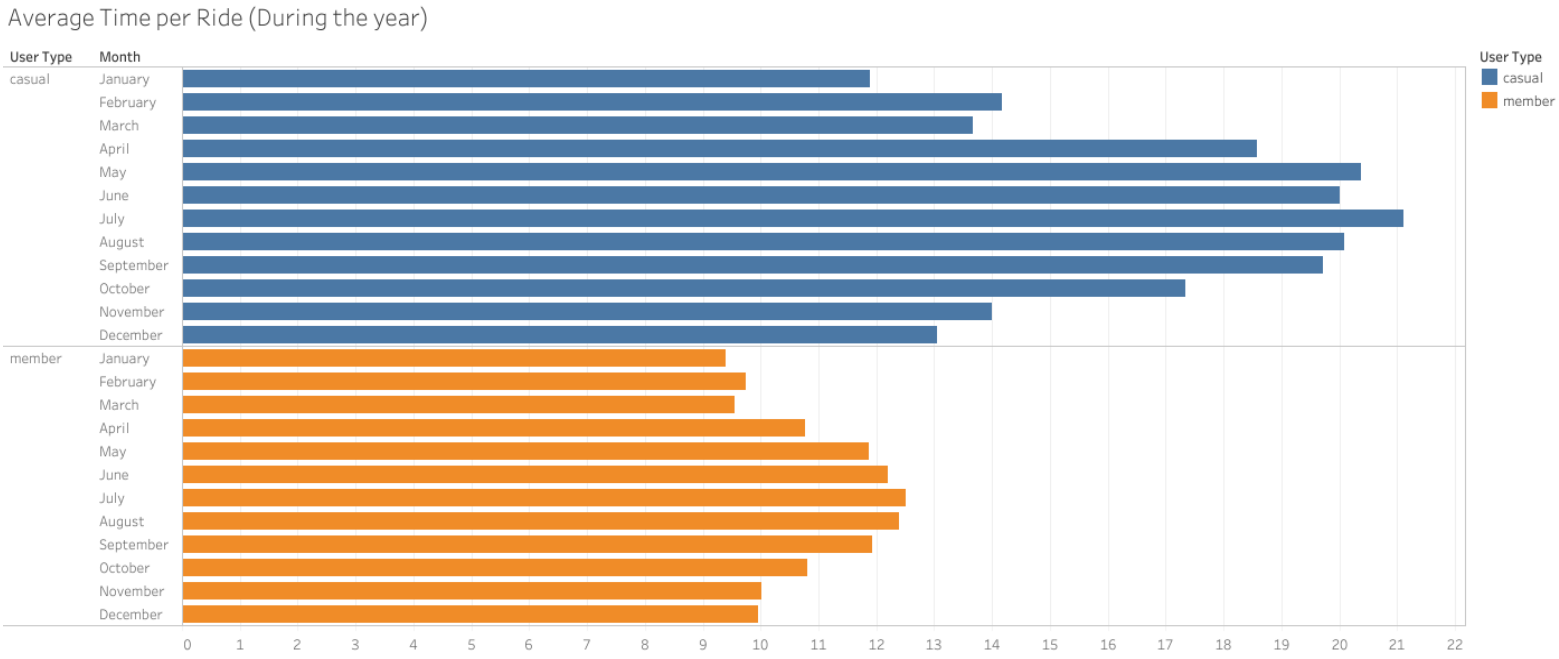
Minutes per Month



Similarly, we can see that in the ‘Minutes per Month’ chart, the summer months are the ones which the users’ time is mostly spent on. Both groups spend their time predominantly in the months from May to September. On the other hand, the winter months have minimal time spent on the service, especially by the Casual users.

The interesting remarks on the matter are to be made when considering the differences in the time between the Casual users and the Members. Generally, Casual users’ habits can be characterized as more ‘extreme’. In the summer months, Casual users spend in total a significant amount of time more, as compared to Members. At the same time, during winter months, they spend significantly less time in total, than the Members group does.

For once more, the conclusion drawn from this chart is that in the total amount of time spent, the Members group shows more ‘consistency’ and far less discrepancy than the Casual users group, throughout the year. The following chart will be shedding some light, more on the personal habits of users, rather than the group as a whole, on this matter.



The above chart describes the average time of a ride throughout the year for the average user of both groups. Even though many more Members use the service during the winter than Casual users, Casual users who ride during the winter months still tend to ride for more time compared to Members.

Again there is impressive consistency throughout the entire year for Members, with each ride being 9 and a half to 12 and a half minutes long, on average. For casual users, once more, the discrepancy is much greater,

with average rides ranging from almost 12 minutes (during the winter) up to 21 minutes (during the summer). Again, it can be concluded that members have a much more ‘homogeneous’ behaviour due to the nature of their use of the service: a means of transportation to and from work, mainly, generally a more ‘streamlined’ use. At the same time, Casual users’ behavior is much more dependent on weather conditions and outside temperatures, since the nature of their use of the service is more casual in essence.

Key Insights from the Data - How do annual members and casual riders use Cyclistic bikes differently?

Comparison: Members vs. Casual Users			
Aspect	**Members (Annual Subscribers)**	**Casual Users (Non-Subscribers)**	**Common Ground**
Total Rides	More rides overall	Fewer rides overall	Both contribute significantly to total usage
Ride Duration	Shorter rides (11-12.5 min avg.)	Longer rides (16-22 min avg.)	Both ride for longer on weekends
Peak Usage Hours	7-9 AM & 5 PM (work commute)	4-8 PM (leisure/recreational)	5 PM is a peak for both groups
Days of Use	Weekdays more popular	Weekends more popular	Both use the service on all days, just with a different emphasis
Seasonal Trends	More consistent year-round usage	High in summer, low in winter	Both prefer the summer months
Bike Type	Balanced use of classic & electric bikes	Slight preference for electric bikes	Both use both bike types frequently

How could the company's strategy be affected by the findings?

As the data has already clearly shown, while each group has its own characteristics and variations, there is a considerable amount of common ground. Based on this, and in combination with the findings, the company's strategy can be oriented towards the integration of the Casual users into their pool of Members. This strategy should consider both the differences and the common ground and offer incentives to Casual users to encourage them to purchase a membership.

Therefore, by considering the most important insights below, we can formulate the appropriate strategy:

- Casual users take longer rides but ride less frequently than members.
- Casual users prefer weekends, while members ride more on weekdays.
- Casual users prefer evening hours, while Members morning and evening hours, with 5 pm being the peak hour for both.
- Casual users are highly influenced by the weather, whereas members ride more consistently year-round, still more during the summer, as well.
- Both groups prefer similar bike types (marketing efforts can focus on other differentiators).

Based on the data, several approaches could be adopted for the membership of Casual users. These approaches will be mentioned in the next part, below. There can be one type of membership with a combination of several of these elements/proposals, or several different types of memberships. This, of course, depends on **Cyclistic's** marketing and sales

planning and convenience. In any case, the solutions proposed below are various and can offer a strong foothold on the market, being based on the actual data of the clients that the company provided.

Proposed strategy points for the conversion of Casual users to Members

- Offer a weekend membership plan for leisure riders
- Offer seasonal memberships for warm-weather riders
- Offer a membership for 4-8 PM for the peak evening hours
- Provide a first-month free trial membership or discounted first 3 months for frequent casual riders.
- Implement a referral rewards system, allowing Members to invite Casual users for discounts.
- Use email campaigns and app notifications to highlight cost savings for frequent casual riders.
- Promote success stories and testimonials of Members and Casual users, highlighting the benefits of cycling, as a means of comfortable transport and an active life.
- Organize exclusive member events (e.g. guided weekend bike tours) with a one-time trial for casual users.
- Introduce a loyalty point system for existing and new Members, or even for Casual riders. Depending on the user, there could be rewards based on their actions or purchases, for example, there

could be points awarded for longer rides or consequent rides for every day, membership renewal etc. The points could offer benefits to the Members (e.g. extending their membership by a free month), or membership offers to the Casual users.

Conclusion

This report focused on the use of data as a source of marketing development, that will in turn increase sales for the **Cyclistic bike-share** Company, by boosting its Members base.

The processed data revealed the habits of both Members and Casual users of the services of **Cyclistic,** pointing towards a strategy of several approaches that could be adopted. The adoption of these approaches will offer:

- Increased flexibility
- Incentives that will make membership financially appealing
- Personalization, based on the behavior of Casual users
- An improved user experience with benefits for continuous use
- Encouragement for using the services and memberships on a long term basis

As always, the results are not always directly visible and experimentation might need to be exercised, but the analysis of data and the subsequent report offered a full insight on the customer needs that **Cyclistic** will need to cover. By having these needs, which align with their habits, covered, the customers will become more dedicated to the company, offering an advantage over competition and consistently enlarging its Members' pool.